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# **How the European Union Fell Out of Love with Open-Source Software**

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# How the European Union Fell Out of Love with Open-Source Software

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## Abstract

In the early 2000s, the American proprietary software company Microsoft dominated the European Union's (EU) markets for desktop and server operating systems, and for productivity software. A coalition of determined open-source software (OSS) advocates and a handful of technology experts working in the European Commission set out in 2004 to end Microsoft's monopoly. They almost succeeded. This article reveals how they managed to change the EU's software policies, made Microsoft lobbyists work overtime - and in the end, and despite their best efforts, could not withstand the power of proprietary companies' lobbying campaigns. Drawing on the Multiple Streams Framework, the article explains the European Commission's decision to promote OSS and open standards in 2004, and its puzzling decision to reverse course just a few years later, in 2010, despite its unchanged rhetoric about the benefits of openness. The analysis reveals three key factors that drove the changes in the EU's policies. In 2004, OSS advocates managed to frame the EU's dependency on proprietary software as a problem – and the promotion of OSS and open standards as the solution. In 2010, Microsoft and other proprietary companies used their existing connections in Brussels to sow doubt about the maturity and cost of OSS among EU policymakers. They also infiltrated open-source expert groups to influence future EU open-source policies. Economic and political developments in technology markets also played in favor of the proprietary incumbents. This article contributes to our understanding of the dynamic interplay between OSS advocacy and the influence of proprietary incumbent companies on EU policymaking. This is important because history is repeating itself with the current tug-of-war over open-source artificial intelligence in Brussels.

**Keywords:** European Union; open-source software; proprietary software; open standards; digital policy; lobbying; multiple streams framework

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## 1. Introduction

When the artificial intelligence (AI) startup Mistral, based in France, entered into a distribution agreement with American “Big Tech”<sup>1</sup> company Microsoft in February 2024, an outcry erupted in Brussels policy circles (Coulter & Chee, 2024). European Union (EU) policymakers were incensed that Mistral had decided not only to sell a stake to Microsoft but also to make its top-performing foundation model<sup>2</sup> proprietary under the agreement with the American technology giant. Previously, Mistral’s executives had championed open-source models, which can be reused by anyone, raising hopes that Europe would finally have its own, independent, highly-performing AI (Burgeot, 2024). Mistral’s decision to partner with American “digital gatekeeper” (European Commission, 2022) Microsoft effectively crushed these hopes. Instead of having its own, independent AI, the EU would remain dependent on American technology giants, at least for the foreseeable future.

The recent discussions among EU policymakers about Mistral, open-source AI, and the EU’s dependency on American Big Tech are reminiscent of a debate that took place in Brussels in the early 2000s. Back then, EU policymakers were not concerned about Europe’s reliance on foreign-made AI, as AI did not play a central role in the digital economy yet. Instead, they fretted about Europe’s dependency on American companies for key software, such as desktop operating systems, server operating systems, and office productivity software (Agarwal, 16 May 2019).

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<sup>1</sup> While there is no agreed-upon definition of “Big Tech,” the expression is often used as an umbrella term for Alphabet (Google’s parent company), Amazon, Microsoft, Meta (Facebook’s parent company), and Apple, i.e., large technology companies that “have amassed tremendous wealth and influence through their ownership of digital infrastructure and platforms” (Khanal et al., 2024, p. 1).

<sup>2</sup> A “foundation model” is a large-scale, pre-trained AI model that serves as a base or foundation for a variety of specific applications and tasks. These models are typically trained on vast datasets and can perform a wide range of functions, such as natural language processing, image recognition, and more.

Microsoft was already the focus of policymakers' worries back then, as the American technology giant dominated the EU market for all these three kinds of software (with its Windows, Windows Server, and Office software, respectively) (Nemati & Barko, 2005, p. 1066). In order to reduce Microsoft's stranglehold on EU markets, policymakers in 2003 and 2004 introduced a number of measures to support open-source software (OSS) and open standards at the expense of the proprietary software sold by Microsoft and other proprietary American firms (Anonymous A). The hope was that public support for open standards and for OSS – the source code of which can be viewed, modified, and shared by anyone – would reduce Microsoft's dominance in the EU. This would be achieved by strengthening Europe's fledgling OSS industry, increasing interoperability among different software products, and thereby decreasing dependency on Microsoft's products.

Most important among the measures introduced by EU policymakers were procurement preferences contained in the European Interoperability Framework (EIF), a comprehensive set of guidelines meant to support the pan-European delivery of e-government services by establishing interoperability principles across various dimensions. These guidelines strongly encouraged EU institutions to buy OSS instead of proprietary software, and to purchase software that implemented open standards, defined as standards with standard-essential patents (SEPs) available for free. Given that the EU's public procurement budget for information and communications technologies (ICT) at the time was worth the Euro equivalent of 50 billion U.S. Dollars (Updegrove, 2010) per year, these procurement preferences had the potential to alter the EU's software landscape and put an end to Microsoft's dominant position in European software markets. However, in a surprising about-face, the EU eliminated the procurement preferences for OSS and changed its definition of open standards just a few years later, namely in 2010. This reversal happened despite Microsoft being just as dominant in EU markets in

2010 as it was in 2004. Today, the usage of proprietary Microsoft software and operating systems in EU public administrations is still extremely common (Maack, 2017).

In this paper, I use an extended version of the multiple streams framework (MSF) originally developed by John Kingdon (1995) to explain (i) why and how OSS advocates were able in the early 2000s to convince the European Commission to promote OSS and open standards against the interests of a company as powerful and well-connected as Microsoft; and why and how the Commission suddenly dropped the preferences for OSS and (truly) open standards in the 2010 version of the EIF. The MSF analytically separates the policymaking process into three streams, namely the “problem,” “politics,” and “policy” streams (Kingdon, 1995). The analytical separation of the policymaking process into these streams allows for structured insights into i) which actors are the most active during each part of the process and ii) how these actors seek to impact the process in order to achieve their preferred outcomes. I also add a “technology stream” (Voß, 2007; Elzen *et al.*, 2011; Goyal *et al.*, 2021) to the MSF in order to show how developments in technology markets empowered large, proprietary incumbent companies and disempowered OSS advocates, contributing to the rollback of pro-OSS and open standards provisions in the EIF.

My contention is that OSS advocates initially succeeded by framing it as an issue in the problem stream that the EU was so dependent on closed-source, foreign-made software – and by suggesting in the politics stream that the solution to the EU’s problem was to use more OSS and open standards. However, by 2010, Microsoft and other proprietary incumbent firms had caught on to the importance of the EU’s procurement guidelines. Incumbent firms therefore i) in the politics stream deployed their significant resources to sow doubt about OSS among EU policymakers and ii) in the policy stream joined expert groups working on the EU’s open-source strategy to ensure that the second version of the EIF reflected their preferences. Microsoft and other proprietary companies were also helped by political and economic

developments in the technology stream, particularly by changes in companies' patenting strategies that emerged in the late 2000s.

My empirical approach relies on both primary and secondary sources. I conducted interviews with 10 OSS policy experts, several of whom were directly involved in the drafting of both versions of the EIF. I also analyzed a large number of official documents published by the European Commission. In addition, I reviewed videos and popular media articles, and I studied several draft versions of the EIF. Examining the drafts was particularly insightful, as it allowed me to trace who tried to shape the EIF – and who succeeded in doing so.

In this paper, I explain why the EU's support for OSS and open standards dissipated between 2004 and 2010 despite the EU's unchanged, unequivocally positive rhetoric about OSS and open standards, and despite the continuing dominance of Microsoft in the EU's software market. This paper therefore deepens our understanding of the dynamic interplay between OSS advocacy and the influence of large incumbent firms on EU policymaking. By applying an extended version of Kingdon's MSF, I dissect the key factors that initially empowered OSS advocates to drive policy changes in favor of OSS and open standards and later led to the reversal of these policies. My research not only provides a thorough analysis of historical policy shifts but also offers valuable insights into the current and future landscape of AI governance in the EU. This is particularly true for governance initiatives relating to the promotion of open-source AI. Ultimately, this study contributes to our knowledge of how the problem, politics, policy, and technology streams interact, shaping the EU's approach to digital sovereignty and its dependency on foreign technology.

## **2. Analyzing and Framing the Dynamics of the EU's Changing Open Source Policies**

The Commission's actions with regard to the promotion of OSS and open standards are particularly surprising because unlike elected governments, which can change every few years, the Commission is a permanent institution with civil servants who ensure policy continuity (Ellinas & Suleiman, 2012). This cadre of civil servants provides stability amid the arrival of new Commissioners (and their cabinets) every five years. It is an intriguing puzzle, then, why the Commission promoted OSS and open standards in 2004 and reversed course just a few years later, in 2010, even though nothing had changed about either the EU's dependency on closed-source, foreign-made software or the Commission's rhetoric about the benefits of openness.

Historical-institutionalism maintains that policy change is typically slow and gradual, and policy options are constrained by previous policies and established institutions (Pierson, 2004). Significant shifts and rapid transformations can occur during "critical junctures," which are preceded by moments of serious crisis or upheaval (Capoccia & Kelemen, 2007). Historical institutionalism thus raises the expectation that policy is "path dependent" (Capoccia & Kelemen, 2007) unless a crisis occurs through which the usual constraints on the policy process are relaxed. While crises and upheaval are important catalysts for policy change, I put forth that unless relevant decisionmakers perceive that there is a concrete problem that can be addressed through a concrete policy solution, no such policy change will occur. This leads me to expect that policy entrepreneurs play a critical role in suggesting relevant policy solutions to perceived problems to decisionmakers.

Thus, in contrast to another prominent theoretical framework for explaining policy change, the Advocacy Coalition Framework (Sabatier, 1988), I hold that the presence of advocacy coalitions promoting particular policy solutions is necessary but not sufficient to effect policy change. In addition to the existence of such coalitions, decisionmakers need to perceive that there is a problem that needs to be solved and policy entrepreneurs need to offer

solutions for solving the problem. The MSF conceptualizes this process in terms of the “coupling” of the problem, politics, and policy streams, which opens a “window of opportunity” during which policy shifts can occur. This raises the expectation that only if all three factors are present, will policy change happen.

When Kingdon (1995) first formulated the MSF, he identified three streams – the problem stream, the politics stream, and the policy stream – that were meant to explain agenda setting in the context of United States (U.S.) politics. Agenda setting is a crucial aspect of the policy process because it determines which issues receive attention and resources from policymakers. Analyzing the agenda-setting process is therefore important for understanding why some problems gain prominence while others are neglected, how political dynamics influence policy priorities, and how solutions are matched with problems at opportune moments. This helps to clarify the mechanisms through which policy agendas are shaped and the factors that drive policy change.

Kingdon’s MSF and its three streams are a useful analytical tool for disentangling the agenda-setting process. Each stream contributes to understanding a particular aspect of the process. In the problem stream, epistemic communities frame societal conditions based on focusing events and indicators, pointing to “problems” that they believe policymakers need to tackle. In the politics stream, “advocacy coalitions” (Kingdon 1995) struggle with each other to match problem definitions with policy tools. Finally, in the policy stream, “instrument constituencies” supply policymakers with information about the design and mechanics of various policy tools. When the three streams are “ripe” for coupling (Kingdon 1995), items enter the policy agenda.

In recent years, many authors (Voß 2007; Elzen *et al.* 2011; Goyal *et al.* 2021) have advocated adding a technology stream to the MSF in order to capture the impact of policymaking on technological innovation and vice versa. The MSF can be used to analyze



policy change in all sorts of policy areas, not just in technology policy. Therefore, the relevance of the technology stream varies. However, in policy areas in which technology (either as a governance tool or as an object of governance) plays an important role, the technology stream can be usefully added to the MSF to shed light on why and how policy change happens. The technology stream models the background activities that contribute to technological innovation, such as research and development (R&D), the creation and protection of intellectual property (IP), the establishment of new businesses, the creation of markets, and technology transfer (Goyal 2019; Goyal *et al.* 2020; Goyal *et al.* 2021, p. 1022). All of these activities are driven by “technology constituencies” (Goyal & Howlett 2018) such as technology manufacturers and suppliers. These constituencies have an interest in supporting technological diffusion and therefore might influence policymaking processes (Raven *et al.* 2016).

## **2.1. Understanding Technology Constituencies**

Technology constituencies are not homogenous. As Goyal *et al.* (2021, p. 1022) point out, “[t]hese constituencies have wide membership – comprising technologists, manufacturers, suppliers, service providers, users, lobby groups, political actors, and academics.” Thus, while all technology constituencies share an interest in the diffusion of new technologies, *which* technologies exactly they want to promote and what governance arrangements they want to advance differs depending on their specific positions and interests.

There are many schisms that can divide technology constituencies. For example, developers of OSS have different views about software patents and about the licensing of standard-essential patents (SEPs) than proprietary software companies (Free Software Foundation Europe (FSFE) 2016; von Ingersleben-Seip & Bütke 2020). Companies that possess a lot of data view data-sharing policies differently than companies that do not hold

large amounts of data (Trampusch 2023). Similarly, companies that have made the protection of personal data one of their value propositions are likely to have different opinions about data protection regulations than companies that have based their business models on exploiting personal data. More generally, incumbent technology companies with strong market positions tend to prefer different policies than emergent technology companies who are fighting for market share. This was very obvious in the case of the EU's open source policies, as will be shown later in this paper.

## **2.2. The Role of Technology Constituencies in Policy Change**

As Goyal *et al.* (2021, p. 1022) point out, the technology stream can affect all three streams of the MSF independently. If the adoption and use of new technologies change prevailing social, economic, and environmental conditions and such changes are widely perceived as negative, this may contribute to the ripening of the problem stream. Public perceptions of new technologies themselves determine whether these technologies are seen as problems or solutions. Thus, a given technology might be framed as a problem that policymakers need to “solve” through regulation or it might be seen in just the opposite way – namely as the solution to an existing problem. In the latter case, technology constituencies can impact the policy stream by positioning the technology as (a key part of) a governance arrangement that is aimed at solving the problem in question. Finally, new technologies can affect the politics stream by bolstering or undermining government or party ideology. They can also support the digitization of governance processes, enabling new actors to enter the politics stream.

Given their different interests, goals, and strategies, incumbent and emergent technology constituencies are likely to play different roles with regard to the three streams. In the early 2000s, OSS advocates did their best to convince EU policymakers that OSS and open standards were the solution to the EU's problem of being too dependent on foreign software.

The American proprietary incumbent companies (particularly Microsoft), on the other hand, tried to persuade policymakers that the promotion of OSS and open standards was misguided for several reasons: OSS was too immature to be deployed for large-scale government projects; the total cost of ownership (TCO) of OSS was higher than commonly assumed; and the adoption of open standards would undermine rather than foster the EU's goal of creating more interoperability.

### **3. Research Design: Tracing the Impact of Proprietary Incumbent Companies and Their Emerging OSS Competitors on the EIF**

The adoption and change of the EIF represent an important case for EU digital policymaking. For one, the EIF has had a significant influence on the EU's interoperability policies for the past 20 years and has served as an important building block of the digital single market (DSM). It was also one of the earliest and most far-reaching policy initiatives aimed at promoting OSS and open standards in the EU. The promotion of OSS and open standards is a goal that is still relevant and important to the EU today (European Commission n.d.-a). Many other organizations in jurisdictions around the world moreover followed suit and implemented pro-OSS policies in the years after the first EIF was published (Lewis 2010; von Ingersleben-Seip & Büthe 2020). Finally, and most importantly, the EIF is a uniquely suitable case to illustrate the different interests, goals, and strategies of proprietary incumbent companies and their emergent OSS competitors, as the Big Tech incumbents and their OSS challengers were deeply divided on what was the problem (or whether there was one at all) and what constituted the best solution to the problem. The adoption of the first EIF in 2004, along with the policy changes made in 2010, offers ample opportunities to study how proprietary incumbents and OSS advocates, respectively, acted in the problem, politics, and policy streams, and how successful they were in their efforts. The case of the EIF is therefore ideally suited for

understanding how Big Tech manages to stifle its smaller competitors and undermine policy initiatives aimed at strengthening the EU's own technology sector.

The data for this study were drawn from (i) 10 interviews with industry representatives and with officials from the European Commission who play(ed) a leading role in shaping policies related to OSS and open standards; (ii) publicly available documents (such as media articles, statements on European Commission websites, press interviews, and videos) in which Commission officials talk about the EIF; (iii) multiple drafts of the second version of the EIF, which show the input of various proprietary incumbent firms and OSS advocacy groups.

I analyzed secondary sources such as media articles, statements on European Commission websites, press interviews, and videos with the help of the software Atlas.ti Cloud, which was used for keyword searches and coding. These sources represented a rich font of information. Additionally, I studied the annotated drafts of the second version of the EIF. Seeing how the drafts changed in accordance with input by different technology constituencies was very valuable for two different reasons. First, it allowed me to see which actors tried to influence the EIF and how they tried to do so. Second, it enabled me to find out whose input was actually taken onboard by the Commission. The second aspect is particularly interesting because it allows for insights into the power of different actors as well as the strategies that work – and do not work – when trying to influence policymaking processes in the European Commission.

#### **4. Explaining the EU's Changing OSS Policies with the Help of the Extended MSF**

Drawing on the extended MSF, I will explain in this section how the actions of both incumbent and emergent technology constituencies in the problem, politics, policy, and technology

streams contributed to the EU's changing OSS policies. First, however, I will provide some context for the adoption of the first EIF in 2004.

#### **4.1. Adopting the EIF to Promote OSS and Open Standards**

In 2004, the software industry looked very different from today. Unlike today, when even Big Tech firms such as Microsoft, Google, and Meta release part of their software under open source licenses, most large software firms in the early 2000s were hostile towards OSS. Instead of distributing OSS for free and then charging for service and maintenance of the software,<sup>3</sup> their business model was to hire programmers who wrote closed source software that was licensed to customers for a fee. Under this proprietary model, source code was closed and could not be studied, modified, or shared by users of the software.

Microsoft, one of the major proprietary software companies at the time, dominated the market for desktop operating systems and for office productivity software, among other things (Nemati & Barko 2005, p. 1066). One of the biggest suppliers of proprietary business and database software, and one of the world's biggest patent holders in 2004, was International Business Machines (IBM) (Shankland 2005). Another business and database software company with many patents to its name was Oracle (United States Patent and Trademark Office (USPTO) 2004). In other words, large proprietary firms headquartered in the United States dominated the software landscape.

In March 2004, Microsoft was fined a half a billion Euros by the EU for abusing its “near-monopoly” position in PC operating systems (Deutsche Welle 2004). OSS development, on the other hand, was not dominated by one large firm (or even a handful of large firms). Instead, it was globally dispersed and mainly driven by small companies or startups, with a lot

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<sup>3</sup> Distributing OSS for free and then charging for service and maintenance of the software was the prevailing business model of OSS back then.

of OSS developers residing in Europe. Unlike today, OSS was still not a mainstream phenomenon. In fact, as my interviewee Peter Dröll (16 May 2019), a high-ranking official from the EU's Directorate General for Research and Innovation (DG RTD), explained, many businesses, governments, and private consumers back then viewed OSS "as risky, as uncontrollable, as probably lower quality – too good to be true." However, governments around the world slowly began to recognize that they could potentially realize a number of economic and political advantages if they deployed more OSS in public administrations and used open standards that were easy to implement in OSS programs.<sup>4</sup>

The use of OSS and open standards has the potential to create a number of benefits for public administrations. Both are enablers of interoperability because they allow any interested party to study them, reuse them, and build upon them. Interoperability in turn may lead to more competition and innovation in software markets because it prevents the emergence of closed, integrated systems (so-called "walled gardens") and thereby counteracts the direct and indirect network effects<sup>5</sup> that drive the emergence of monopolies in these markets. For governments aiming to create a more dynamic domestic software industry, the prospect of increasing competition and innovation through public institutions' use of OSS and open standards surely seems very attractive. In the specific case of the EU, interoperability was also important for harmonizing the ICT systems of the different member states and laying the foundation for the DSM. Additionally, all governments may appreciate the fact that the openness and accessibility of OSS code affords them a measure of control over the software they use and enables the transfer of high value knowledge to local developers studying the code (Rajani et al., 2003).

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<sup>4</sup> For example, the Commonwealth of Massachusetts in the United States announced a state-level initiative in 2003 that required all information technology expenditures in 2004 and 2005 to be made on OSS (Lettice 2003).

<sup>5</sup> Direct network effects occur when an additional user makes a product or service more valuable to existing users. Indirect network effects are market-mediated and occur when the increased availability and decreased price of complementary goods make a product or service more valuable to existing users.

### *Framing American Proprietary Software Companies' Market Dominance as a Problem*

The fact that the EU's ICT market was dominated by large American companies in the early 2000s (one of which – Microsoft – had moreover recently been found to abuse its near-monopoly position) was framed by OSS advocates inside and outside of the European Commission as a problem that needed to be addressed by policymakers. Promoting the use of OSS and open standards was offered as a solution. Specifically, the leaders of the EU's Directorate General for Informatics (DG DIGIT) and the Directorate General for Enterprise and Industry (DG ENTR)<sup>6</sup> made the case that it was important to reduce the EU's dependency on American software companies, primarily to strengthen the EU's own ICT industry (Anonymous A, 21 May 2019), promote competition between foreign and European firms (Updegrove 2010), and lower the risk that software made outside of the EU could be used as instrument of statecraft by foreign powers (Brunet, quoted in Gold 2015). The leaders of DG DIGIT and DG ENTR had the backing of newly-elected EU Competition Commissioner Neelie Kroes, who strongly supported the uptake of OSS and open standards within EU public administrations during her tenure (Fiveash, 2008; Tait, 2008), both in order to save costs and in order to safeguard the EU's digital autonomy. The ICT experts within DG DIGIT and DG ENTR, too, were aligned with the DGs' leadership because they viewed OSS, supported by open standards, as more cost-effective, secure, and transparent than proprietary software (de Vriendt, 27 May 2020). In addition, OSS activists such as Richard Stallman and the Free Software Foundation (FSF) were vocal in framing the use of closed source software in general as a problem, pointing to spying incidents such as the special access codes of the U.S. National Security Agency (NSA) that had secretly been built into the Microsoft Windows operating system and were discovered by cybersecurity researchers in 1999 (Campbell 1999). After EU

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<sup>6</sup> Note that the Directorate General for Enterprise and Industry (DG ENTR) no longer exists, as it was replaced on 1 January 2015 with the Directorate General for Internal Market, Industry, Entrepreneurship, and Small and Medium Enterprises (DG GROW).

public administrations had been comfortable for years with buying and using Microsoft's closed source software, OSS advocates' efforts to frame the use of proprietary American software as a problem prompted European policymakers to actively try to increase the use of OSS and open standards in the EU. If EU policymakers didn't believe the framing of the OSS advocates, it would have been easiest for them to simply buy software from another proprietary American vendor.<sup>7</sup> Considering this "implicit counterfactual" (Fearon, 1991, p. 174) strengthens the hypothesis that it was OSS advocates' framing of the EU's dependence on proprietary software as a problem that convinced EU policymakers to promote OSS and open standards.

#### *The European Commission as a Policy Entrepreneur*

The technical staff of DG DIGIT and DG ENTR then played the role of policy entrepreneurs in the policy stream by drafting the "European Interoperability Framework for pan-European eGovernment<sup>8</sup> Services" (colloquially known as "the first EIF"), a document that laid out in detail the EU's interoperability strategy and the role of OSS and open standards within it. The first EIF advised EU public administrations to use OSS and open standards to increase the technical interoperability of the e-government services of the EU and its member states. The Commission seemed intent on leveling the playing field between proprietary and OSS companies by recommending that public administrations in the EU consider OSS favorably alongside proprietary software and moreover buy software implementing open standards (European Communities 2004, p.10). In effect, this meant that EU public administrations were

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<sup>7</sup> The reason this would have been easier than trying to switch to OSS is that, at the time, OSS wasn't as user-friendly as proprietary software and therefore a lot more effort and technological knowledge was required to use it.

<sup>8</sup> The European Commission uses the spelling "eGovernment," while I use the spelling "e-government" in this paper.



likely to give preference to OSS over proprietary software in their public procurement decisions in the future – a significant development given the EU’s large public procurement budget.

What was even more remarkable about the first EIF was the way in which open standards were defined in the document. The Commission’s definition of an open standard required owners of SEPs to license these patents on a royalty-free basis in order for the standard to be considered open. This was welcomed by open source advocates, who pointed out that royalty-free licensing of SEPs would enable the use of open standards in all kinds of software, even OSS licensed under the highly permissive General Public License (GPL).<sup>9</sup> The definition was, however, disadvantageous for SEP holders, as it precluded them from monetizing these patents, even under a fair, reasonable, and non-discriminatory (FRAND) license. Without such policy entrepreneurship, it is unlikely that the EU would have veered off its previous path, especially since historical institutionalist scholars have shown that path dependence often keeps in place suboptimal technologies or technological arrangements (Dobusch & Schüßler, 2013). The historical institutionalist argument seems highly applicable in the case of OSS, as OSS was a new technology that was surrounded by a lot of uncertainty (Dröll, 16 May 2019) and that came with high switching costs because of its lack of simple graphical user interfaces.

#### *The Technology Stream’s Contribution to Understanding the Actions of the Incumbents*

Notwithstanding the disadvantages the first EIF created for SEP holders, there was not a large lobbying effort to change the text of the EIF when it was in the drafting phase (de Vriendt, 27 May 2020). A look at what was happening in the technology stream helps to explain this. In the early 2000s, most large software and hardware players owned SEPs. Therefore, the industry was not much concerned with negotiating payment terms for SEP licenses (Kramler, 16 May

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<sup>9</sup> The GPL does not allow any restrictions on the distribution of software, meaning that, for example, royalty payments based on the number of distributed copies of a certain software program are incompatible with the GPL.

2019). Instead, firms simply licensed their SEPs to each other, employing a cross-licensing model that made payments unnecessary. OSS, as mentioned above, was still a marginal phenomenon that was mainly driven by individual developers interested in attaining reputational benefits (Lakhani & Wolf 2003); thus, there were not a whole lot of OSS companies yet. Therefore, SEP holders were not much concerned about having to give away their intellectual property for free on a large scale. Instead, they expected to get other SEP holders' patents in return for their own patents, which meant that they did not view the text of the first EIF as especially threatening (Kramler, 16 May 2019). Also, since the first EIF was a novel kind of policy (the EU had not published any interoperability frameworks before), a lot of firms did not realize the significance of what DG DIGIT and DG ENTR were working on until after the text had been published and its ramifications became clear.

In 2004, there was therefore a “window of opportunity” (Kingdon, 1995) that policy entrepreneurs within DG DIGIT and DG ENTR exploited. Officials at the highest levels of the Commission as well as the leaders of DG DIGIT and DG ENTR were aligned on promoting OSS and open standards to the largest extent possible. This allowed the technologists within the two DGs, who viewed OSS as superior to proprietary software for many reasons,<sup>10</sup> to draft a text that contained far-reaching provisions in support of OSS and of open standards, defined in a way that allowed even OSS developers releasing their software under the GPL to implement these standards in their products. Proprietary companies were not very concerned about the ramifications of the EIF because of the prevailing market conditions in 2004 and because of their failure to understand the potentially far-reaching consequences of the framework. This enabled the European Commission to publish guidelines that advised EU public administrations to give preference to OSS over proprietary software in public

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<sup>10</sup> These reasons included the cost-effectiveness, transparency, and security fostered by OSS (de Vriendt, 27 May 2020).

procurement and that forbade the monetization of SEPs, an important step towards nurturing and strengthening the EU's OSS industry. The exploitation of this window of opportunity by policy entrepreneurs within the Commission constitutes a key element of my explanation of outcomes.

#### **4.2. Adopting the Second Version of the EIF**

In 2010, the European Commission published the second version of the EIF. In the updated guidelines, the definition of open standards – which were now called “open specifications” – had changed. The Commission no longer asserted that any intellectual property included in a standard needed to be licensed on a royalty-free basis in order for the standard to be considered open. Instead, it stated that licensing SEPs on FRAND terms was also acceptable: “If the openness principle is applied in full [...], [i]ntellectual property rights related to the specification are licensed on FRAND terms or on a royalty-free basis in a way that allows implementation in both proprietary and open source [sic] software” (European Commission 2010, p.26). The Commission did not address the incompatibilities between FRAND licenses and permissive OSS licenses such as the GPL.

This seemingly counterintuitive move by the Commission created an uneven rather than an even playing field for OSS and proprietary software, raising questions about the driving forces behind the changes to the second version of the EIF. The variation in the two versions of the EIF is highly surprising for three reasons: a) The Commission is to a large degree composed of civil servants, which should foster a high level of policy continuity (Ellinas and Suleiman, 2012); b) the Commission did not change its (very positive) rhetoric about OSS and open standards between 2004 and 2010; and c) the EU software market was still dominated by Microsoft and other American proprietary companies in 2010. However, the drivers of the

Commission's decision become much clearer when analyzing what was happening in the problem, politics, policy, and technology streams.

The problem that had been identified by DG DIGIT and DG ENTR in 2004, and that had driven the adoption of the first EIF in November 2004, had not changed. While the EU had certainly made strides in adopting OSS and open standards internally (European Commission n.d.-b), there were still many good reasons to continue promoting OSS and open standards through public procurement initiatives. These included reducing lock-in and the dependency on foreign software suppliers, reducing costs, and strengthening the EU's home-grown ICT industry. Competition Commissioner Kroes also still strongly supported increasing the use of OSS and open standards among EU governments and businesses, saying that adopting open standards was "a smart business decision" (Kanter 2008). The technology experts in DG DIGIT and DG ENTR were also still convinced that the EU needed to lower the cost of its ICT spending and improve the accessibility and transparency of its ICT systems – and that using more OSS and open standards was vital for achieving these goals (de Vriendt, 21 May 2019).

### *Spreading Fear, Uncertainty, and Doubt: Incumbent Technology Constituencies' Activities in the Politics Stream*

However, unlike in 2004, the drafting process of the new version of the EIF was highly politicized. In the politics stream, incumbent technology constituencies – that is to say, proprietary companies, their lobbyists, and a range of non-profits supported financially by proprietary companies – worked hard to convince the European Commission that promoting the use of OSS and open standards (especially open standards as defined in the first EIF) was *not* the solution to the problem the Commission had identified. Rather, these lobbyists claimed, promoting OSS and open standards would actually *create* all sorts of problems for European citizens and governments. Karel de Vriendt, a former European Commission official who had

been directly involved in “the conception and promotion” (de Vriendt, 2017) of the EIF stated that:

The preparation of the second version [...] proved to be a long and difficult process, during which powerful lobby groups did everything they could to weaken the text on “open standards.” In the final text of the second version, although the spirit of open standards remained, the wording was very much less clear and open to different implementations (and misuse). (de Vriendt, 2017)

Specifically, incumbent technology companies sought to convince policymakers of four points: First, that OSS was not actually cheaper than proprietary software when looking at the total cost of ownership (TCO). Second, that keeping the definition of open standards that had been used in the first version of the EIF would lead to EU public administrations losing access to widely-used technologies such as WiFi or GSM. Third, that OSS was not mature enough to be deployed for large-scale and / or mission-critical e-government projects. Fourth, that business model neutrality (i.e. dropping the preference for OSS and open standards that was included in the first EIF) would lead to public administrations getting the best possible ICT solutions for their needs at the best price.

Thomas Kramler, the official from DG COMP, confirmed that proprietary companies were keen to point out that OSS was not necessarily cheaper overall than proprietary software:

There were a lot of arguments I remember about the cost of ownership. For instance, they were saying: “If you look at, overall, what it’s going to cost you to switch to open at this time...” There were also discussions about [...] some member states, or even

some towns specifically, to switch to Linux, like the Munich project at the time. And some others were also contemplating it. There was quite some fear also on the other side, I think, that there would be a big inroad of open source among public institutions, public administrations, which didn't materialize in the end. But the arguments were made – I think especially Microsoft was quite forceful at the time – that open source was going to be a big competitor of theirs, which has not materialized as such. I think they have probably seen the light on open source. But at the time, there were a lot of these arguments on cost of ownership. (Kramler, 16 May 2019)

The Association for Competitive Technology (ACT), which at the time counted Microsoft and big SEP holder American Telephone & Telegraph Company (AT&T) among its sponsor members, meanwhile lauded the Commission for defining open standards in a way in the final draft of the second EIF that allowed for the FRAND licensing of SEPs:

This allowance for some *less* open solutions [i.e., standards whose SEPs are licensed on FRAND terms rather than given away royalty-free] presumably will allow European governments to continue using the hundreds of international open standards that they currently use, which do not meet the open standard definition created by the EIF. This is an important step toward making the EIF a more practical document. Without it, European governments could have been forced to immediately replace technologies like GSM, WiFi, Bluetooth, and DVDs that are widely distributed throughout European citizens, businesses, and governments. This would undoubtedly be incredibly expensive and lead to less interoperability rather than more as business and citizens decided whether to update their existing systems or buy new ones based on government procurement decisions. (Association for Competitive Technology (ACT) n.d., p. 7)

American company General Electric (not primarily a software company but a large SEP holder) asserted in 2009 that OSS was only suited for “playground” applications used internally by companies, putting into question the maturity and safety of OSS and therefore its fitness for large-scale, mission-critical e-government initiatives (Schestowitz, 2009).

At the same time, the Business Software Alliance (BSA), a group that counted Microsoft, SAP, IBM, Dell, and Hewlett Packard (HP) among its 80 industry members at the time, stated that “[w]hen open source is the best solution, the most cost-effective and most suited solution to a given governments [sic] needs then that is of course what a government should procure. But we’re against preference policies” (Müller, quoted in Fiveash, 2008).

As Goyal et al. (2021) point out, in the context of emerging technologies, finding regulatory alternatives “that meet selection criteria such as technical feasibility, financial viability, and value acceptability – can be especially challenging.” Incumbent technology constituencies took advantage of this fact by framing OSS as immature and unsafe and by asserting that defining open standards as royalty-free standards would actually *undermine* the EU’s attempts to create more interoperability. Without the fear, uncertainty and doubt spread by the incumbents in the politics stream, it is hard to imagine that the European Commission would have altered its OSS and open standards policies despite the unchanged market conditions. Thus, I assert that these activities are a key element in explaining the European Commission’s decision to change the definition of open standards in the second version of the EIF to allow for the FRAND licensing of SEPs, enabling SEP holders to monetize their patents.

*Denouncing Incumbents’ Lobbying: Emergent Technology Constituencies’ Activities in the Politics Stream*

OSS advocates were much less active in the politics stream than incumbent technology constituencies. In fact, rather than concentrating their efforts on reiterating that OSS and open standards were the right solutions for the problems identified by the European Commission, the open-source community mainly spent its resources on investigating, analyzing, and complaining about the lobbying efforts of the incumbent constituencies. Both the FSFE and the Open Forum Europe (OFE) attributed the changed definition of open standards in the second version of the EIF to “a pretty intense wave of lobbying in Brussels” (OFE, n.d.). The FSFE (2010) also presented evidence for this assertion on its website, where it shared a document in which it compared an initial draft of the second version of the EIF to a later draft. In the later draft, the Commission had adopted many changes suggested by the BSA, the interest group representing developers of proprietary software such as Microsoft and Oracle. This was especially true for the BSA’s suggestions regarding procurement preferences, competition among different standards, and the definition of open standards itself.

Open-source companies and activists also wrote a number of blog posts and media articles denouncing the lobbying practices of their proprietary rivals (e.g., Bohannon 2010; Diedrich 2010; Schestowitz 2010). Additionally, they spoke to media outlets, especially OSS-friendly outlets such as *The Register* and *ZDNet*, to expose and denounce the advocacy campaigns of Microsoft and others. However, while the open-source community’s efforts were certainly helpful in revealing the nature and extent of lobbying by incumbent technology constituencies, particularly proprietary American software companies and their proxies, they ultimately failed to convince the European Commission that it needed to retain its original definition of open standards in the second version of the EIF.

There were several factors driving the lack of success of the emergent open-source community’s lobbying efforts. According to resource dependence theory, private actors’ ability to engage in political activities and sustain such activities over time is contingent upon the



financial means and expertise these actors possess (Bouwen, 2002; Lowery, 2007; Binderkrantz et al., 2015; Rasmussen & Gross, 2015; Hanegraaff et al., 2020). Monetary resources are necessary to launch and sustain advocacy efforts while expert knowledge helps to gain the attention of policymakers and shapes calculations about the costs and benefits of particular policy options. It is not surprising that the large corporates making proprietary software had more financial resources at their disposal than the open-source community (which consisted mainly of smaller firms, individual developers, and unpaid activists passionate about the cause). Proprietary software firms therefore could spend more on their lobbying campaigns,<sup>11</sup> which meant more meetings with decisionmakers, more articles in the press, more speaker slots at important events, and so on. In terms of expertise, it is unclear whether proprietary lobbyists and OSS advocates were equally well qualified. However, what is clear is that proprietary companies and their lobbyists had more established relationships with decisionmakers than OSS activists and it was therefore easier for them to access the right audience for their arguments.

For example, according to a document shared on WikiLeaks (Anonymous, 2009), lobbyists with ties to Microsoft used their status and credentials as technology specialists to join the European Commission's expert group on OSS in 2008 and subsequently tried to alter the content of the group's European software strategy report. The emergent OSS technology constituency, on the other hand, mainly spoke to – and through – blogs and small media outlets that were passionate about promoting open source and open standards. These blogs and media outlets were, however, so specialized that they were unlikely to reach any policymakers who weren't already very interested in OSS and open standards. Proprietary software companies, in contrast, had dedicated communications officers (both within the companies themselves and

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<sup>11</sup> For example, Microsoft in 2009 and 2010 spent a combined 10 million Euros on lobbying the European Commission (European Union, n.d.a).

within lobbying organizations sponsored by the companies) that had contacts in all the world's largest media outlets and could therefore fairly easily place quotes and articles reflecting their viewpoints within major, widely-read publications. By influencing media discussions, proprietary companies were able to counter the narrative from the emerging OSS constituency. This narrative advocated for the promotion of OSS and open standards as the optimal solution for reducing ICT costs in the EU, strengthening the EU's ICT industry, and decreasing Europe's dependency on foreign software companies.

*Using Their Access to Policymakers to Change Key Passages of the Draft EIF: Incumbent Constituencies' Activities in the Policy Stream*

During the drafting process of the second version of the EIF, incumbent technology constituencies were highly involved in commenting on, and altering, the draft framework. Many diverse groups (such as ACT and the BSA on the side of the incumbent tech constituencies, and the FSFE and the OFE on the side of the emergent tech constituencies), as well as individual companies (such as U.S. cybersecurity company Symantec) and national standards organizations (such as the Lithuanian Standards Board), answered the European Commission's call for comments on the July 2008 draft of the second EIF. The groups responding to the call submitted detailed evaluations of the draft version of the EIF and delivered specific suggestions for improvement.

These suggestions directly influenced the draft version (and therefore the final outcome) of the second EIF, as is shown by a thorough analysis conducted by two FSFE researchers (Roy & Gerloff, 2009). The FSFE document compares side-by-side the consultation draft of the second EIF released in July 2008, the comments submitted by the BSA in response to the consultation draft, a leaked draft from November 2009, and a further leaked draft from March 2010. The comparison of the July 2008 and November 2009 drafts shows

that the Commission incorporated in the November 2009 draft many of the suggestions made by the BSA. The BSA and other members of the incumbent tech constituency thus successfully used a strategy of shaping the exact wording of the second version of the EIF in order to ensure that it would not undermine proprietary software companies' ability to monetize their SEPs and that it would not include public procurement preferences for OSS.

Proprietary companies and their lobbyists were able to exploit their existing connections with decisionmakers when it came to shaping the draft version of the second EIF. Thus, both ACT and the BSA commented on the July 2008 draft version *before* it was officially released for public consultation (Fiveash, 2008), firing the opening salvo in the fight over the second EIF before others had a chance to submit their comments. This shows that the large proprietary incumbent firms enjoyed privileged access to EU decisionmakers and to draft versions of important ICT policy documents that members of the emergent OSS technology constituency simply did not have.

#### *Trying to Safeguard Openness Provisions in the EIF Through Public Comments: Emergent Technology Companies' Activities in the Policy Stream*

Emergent OSS technology constituencies submitted comments on the public, July 2008 draft version of the second EIF. With their input, OSS advocates aimed to safeguard both the definition of open standards and the procurement preferences for OSS that had been included in the first EIF. In parallel, OSS activists pointed out on their own websites and blogs, as well as in friendly media outlets, why a changed definition of open standards that allowed for FRAND licensing of SEPs was problematic for the EU's plans to promote the uptake of OSS.

Thus, the FSFE (2010) wrote that the FRAND licensing of SEPs “clashes with most common Free Software licenses, which forbid restrictions on distribution.”<sup>12</sup> The OFE (n.d.), claimed that allowing patent owners to license SEPs on FRAND terms “effectively locks out open source.” One incompatibility was, the OSS activists explained, that individual FRAND licenses could not automatically be transferred to third parties as required by the GPL (FSFE 2016). Ultimately, what many open-source advocates wanted was a return to the formulation used in the first version of the EIF, which stated that all patents included in a standard had to be made available on a royalty-free basis for the standard to be considered open.

Incumbent technology companies used their resources and expertise to immediately respond to the open-source community’s arguments and try to debunk them. Thus, ACT asserted in comments about the final draft of the second EIF that:

Creating an environment where companies are willing to share their best technology with their competitors requires a careful balancing act. In some cases, these companies can and will agree to Royalty Free Non Discriminatory Licenses for the technologies involved in the standard. In other cases, however, companies have invested millions in developing the technologies and their shareholders are looking for some return on that investment. [...] Without any possibility for royalties, companies will be much less likely to bring their best innovations to the table, jeopardizing the standards process.

(ACT, n.d., p. 7)

Similarly, the BSA wrote in comments on the final draft of the second EIF that the framework’s definition of open standards, which required royalty-free licensing, would force companies to

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<sup>12</sup> As Mitchell and Mason (2011, 32) explain: “Under the FRAND Licensing model, it will usually be the case that although each licensee will be able to obtain his own individual license from the patent owner, the process does not have the same automaticity in respect of downstream users as is involved in the GPL Licensing model.”

give up their patents and relinquish any royalties (Updegrove, 2010). The BSA then urged the Commission “to ensure that innovators who own patents and other intellectual property (IP) can participate in the procurement of eGovernment services in Europe” (EURACTIV, 2010).

In summary, while the emergent OSS technology constituency certainly commented both on the draft versions of the second EIF and the lobbying efforts of proprietary companies, Big Tech’s better connections with policymakers and larger resources were two advantages that the OSS community ultimately could not beat. For one, the propriety incumbents had better access to relevant decisionmakers and therefore to early drafts of the second version of the EIF, which allowed them to frame the narrative from the start by asserting that forcing companies to give away their SEPs for free would undermine innovation. Second, the incumbents’ large financial reserves enabled them to retain many lobbyists who immediately responded to any claims made by the OSS community, which helped to invalidate the OSS community’s arguments in the eyes of policymakers.

In combination, the incumbent technology constituency’s activities in the politics and policy streams had the intended effect. In 2009, when the technocratic experts at DG DIGIT and DG ENTR were in the process of drafting the second version of the EIF, the support of DG DIGIT’s Director General for their work gradually faded as a result of “heavy pressure from outside sources” (de Vriendt, 27 May 2020). This caused a “constant fight” between the technocratic experts and the Director General and, ultimately, a significant weakening of the provisions supporting OSS and open standards (de Vriendt, 27 May 2020).

### *The Technology Stream and Its Influence on the Politics and Policy Streams*

Analyzing what happened in the technology stream provides further important insights on why incumbent and emergent technology companies behaved the way they did. For one, as mentioned above, the big SEP holders at the time of the first EIF were not very concerned

about having to license their patents on a royalty-free basis because they tended to employ a cross-licensing model that made royalty payments unnecessary. In the early 2000s, most large software and hardware players owned SEPs. Therefore, the industry did not care much about negotiating payment terms for SEP licenses. Instead, firms simply licensed their SEPs to each other, employing a cross-licensing model that made payments superfluous. However, by 2009, when the second EIF was being drafted, the licensing of SEPs became more important because the equilibrium created by the cross-licensing model that had been common in 2004 had been upset by the recent market entry of several players that did not own SEPs. SEP holders therefore started to care about the kinds and height of royalty payments they would receive for their SEPs (Kramler, 16 May 2019).

Perhaps even more importantly, by 2009 many large *European* SEP holders had become involved in the debate about the definition of open standards, arguing that SEPs contained in a standard did not have to be given away royalty-free in order for the standard to qualify as open. According to both Kramler (16 May 2019) and an industry representative speaking for a trade association that represents numerous large SEP holders (Anonymous C, 17 May 2019), many firms headquartered in EU member states – including Nokia (based in Finland), Ericsson (based in Sweden), Philipps (based in the Netherlands), and Technicolor (based in France) – were among the SEP holders making these arguments, which meant that the Commission was inclined to listen to the arguments.

Additionally, in the years between 2004 and 2009, OSS and open standards had gained significant momentum in Europe and around the world. Internally, the European Commission was switching to the open-source operating system Linux as server operating system and the open-source webserver software Apache for the Europa.eu server. It also began using OSS for its blogs and public forums (European Commission, n.d.a). And externally, several countries and cities were promoting the uptake of OSS and open standards as well: For example, the

German city of Munich started migrating all of its desktop computers from Microsoft Windows to Linux in 2004. The French parliament passed a law in 2005 that required public administrations to use OSS whenever possible. And the Spanish government followed suit in 2007, passing a law that mandated the use of open standards in all government IT projects. There were other government initiatives as well, e.g., in Italy and Brazil.

This led to the incumbent proprietary companies fearing that a big move towards OSS and open standards was underway in public administrations and public institutions globally (Kramler, 16 May 2019). If the incumbent technology constituency's fears had indeed come true (which they to a large extent did not, thanks in part to the incumbents' lobbying efforts), this would have had serious financial consequences. Thus, the second EIF was part of an ICT strategy that influenced how the EU was going to ultimately spend "up to the equivalent of \$50 billion a year in public IT procurement funds" (Updegrove, 2010). Therefore, while the incumbent proprietary companies had viewed early OSS advocates as heretics and firebrands that could stir up some controversy but were unlikely to change the structure of the ICT industry, they now realized that OSS companies could take away significant business from them. They therefore doubled down on their lobbying efforts to contain the OSS threat while at the same time learning about OSS internally in order to see how it could benefit them (Vaughan-Nichols, 2020). In the case of the second EIF, the incumbents' lobbying campaigns were successful, as they led to the OSS and open standards provisions in the EIF being significantly watered down.

## **5. Conclusion**

The current discussions in the EU about open-source AI mirror the debates about OSS taking place in Brussels in the early 2000s. Then, as now, the EU was dependent on proprietary American digital technologies. Drawing on an extended version of the Multiple Streams

Framework, I have shown in this paper how OSS advocates framed this dependency as a problem and the promotion of OSS and open standards as the solution. I have also argued that the success of their efforts, specifically the promotion of OSS and open standards in the first EIF, brought Big Tech lobbyists to the scene. These lobbyists spread fear, uncertainty, and doubt about OSS among their existing connections in Brussels and influenced the EU's OSS policies by infiltrating expert groups responsible for formulating the EU's future open-source strategies. American incumbent firms were also helped by the fact that European SEP holders became involved in the discussions about the definition of open standards in the second version of the EIF, advocating for the proprietary incumbents. In the second version of the EIF, the EU significantly weakened the provisions promoting OSS and open standards. In 2017, a team of investigative journalists looking into the use of proprietary software in the EU found that the usage of Microsoft software and operating systems in EU public administrations was still "extremely widespread" (Maack, 2017). The journalists concluded that "[all] across Europe, from Finland to Portugal, Ireland to Greece, governments rely on Microsoft software" (Maack, 2017). And a report published by EURACTIV in 2025 revealed that "European Commission officials fear that their heavy reliance on Microsoft constitutes a clear breach of EU data rules" (Wold, 2025).

My paper makes three contributions to the literature. First, it sheds light on the interactions between emergent and incumbent technology companies in the problem, politics, policy, and technology streams. By doing so, I answer a recent call by Nihit Goyal et al. (2021, p. 1029) to "examine different types of interactions among incumbent and emergent technology companies in the technology stream and study how these dynamics influence the policy-making streams and technology policy co-evolution." Second, I explain a significant and puzzling shift in EU policymaking that has had a lasting impact on the EU's software industry, with effects still felt today. Third, by unraveling the struggle between open-source advocates



and proprietary incumbent firms over OSS in the early 2000s, I help to put the current political discussions and battles about open-source AI into context. By providing context, I contribute to a better understanding of the causes and consequences of the political decisions being made with regard to open-source AI today.

Future research can add to my analysis in at least two ways. First, by looking backwards once more. Thus, it would be interesting to find out what happened in the years between the second EIF, published in 2010, and the third EIF, published in 2017. The third version of the EIF in many ways represents a compromise between the interests of OSS advocates and the interests of proprietary incumbents. Therefore, it would be valuable to analyze the actions of both incumbent and emergent OSS constituencies in the political streams and in the technology stream in the intervening years. Doing so would allow for insights into what kinds of strategies and behaviors drove the compromise outcome in the third EIF. Second, it would be interesting to gain a better understanding of what is currently happening in the EU with regard to open-source AI. Especially relevant are attempts by proprietary AI companies to make their AI systems seem more “open” than they really are (Widder et al., 2023). In particular, it would be fruitful to research whether OSS advocates and Big Tech companies are deploying the same strategies in the political streams as the incumbent proprietary companies did in the early 2000s and whether, and if so how, their efforts are impacted by developments in the technology stream.

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## APPENDIX A: LIST OF EXPERT INTERVIEWS

| Interviewee          | Nationality | Type of Role                                                                     | Type of Organization                | Sector     | Mode and Date          | Duration   |
|----------------------|-------------|----------------------------------------------------------------------------------|-------------------------------------|------------|------------------------|------------|
| Agarwal, Prabat      | Indian      | Deputy Head of Unit, DG CNECT (Communications Networks, Content, and Technology) | European Commission                 | Public     | In-person, 16 May 2019 | 45 minutes |
| Anania, Loretta      | Italy       | Program Officer of Net Innovation Unit, DG CNECT                                 | European Commission                 | Public     | In-person, 15 May 2019 | 71 minutes |
| Anonymous A          | Belgium     | Civil Servant                                                                    | European Commission                 | Public     | Online, 21 May 2019    | 41 minutes |
| Anonymous B          | Italy       | Civil Servant                                                                    | European Commission                 | Public     | Online, 24 May 2019    | 42 minutes |
| Anonymous C          | France      | Executive                                                                        | International ICT Trade Association | Non-profit | In-person, 17 May 2019 | 39 minutes |
| de Vriendt, Karel    | Belgium     | Board Member                                                                     | OFE (Open Forum Europe)             | Non-profit | Online, 26 May 2020    | 69 minutes |
| Dröll, Peter         | Germany     | Director of Industrial Technologies Unit, DG RTD (Research & Innovation)         | European Commission                 | Public     | In-person, 16 May 2019 | 31 minutes |
| Kramler, Thomas      | Austria     | Head of Digital Single Market Taskforce, DG COMP (Competition)                   | European Commission                 | Public     | In-person, 16 May 2019 | 34 minutes |
| Lobo, Georges        | Portugal    | Program Officer, Interoperability Unit, DG DIGIT (Digital Services)              | European Commission                 | Public     | Online, 23 May 2019    | 46 minutes |
| Przybyszewski, Marek | Poland      | Project Manager, DG DIGIT                                                        | European Commission                 | Public     | In-person, 17 May 2019 | 39 minutes |

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